

Krishnaa Vadivel

Embedded Systems Engineer | Firmware and FPGA

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Portfolio: <https://krishnaa423.github.io/personal-website/> | GitHub: <https://github.com/krishnaa423>

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Summary

Embedded and systems engineer with hands-on experience across MCU firmware, Lattice FPGA logic, sensor logging, board-level debugging, GPU-enabled diagnostics, and companion engineering software.

Technical Skills

Embedded/hardware: C, C++, Verilog, Lattice FPGA, PIC32, dsPIC, STM32, C8051, sensor interfaces, oscilloscope debugging, Altium

Software: CUDA, OpenCL, OpenGL, Python, C#, ASP.NET, Microsoft SQL Server, Linux

Modeling: DOLFINx, PETSc, finite-element analysis, numerical Python

Experience

Software and Embedded Systems Engineer

Probe Technologies / Weatherford Wireline Services, Houston, TX

2019–2021

- Implemented sensing, logging, and control functionality across Lattice FPGA, PIC32, dsPIC, STM32, and C8051 platforms.
- Investigated hardware/software faults with oscilloscope-based validation, system integration tests, field debugging, and targeted Altium PCB updates.
- Delivered CUDA/OpenCL waveform diagnostics and OpenGL visualization software used alongside downhole instrumentation.
- Built C#/ASP.NET/SQL Server applications for remote test control, sensor calibration, and engineering-data workflows.

PhD Researcher

Yale University, Electrical Engineering

2021–present

- Apply scientific-computing and finite-element methods to semiconductor and instrumentation-adjacent modeling problems.
- Support hardware-oriented teaching and lab work alongside GPU/HPC software development.

Selected Projects

Matrix-multiply accelerator — Implemented a Verilog matrix-multiply accelerator to practice hardware-aware AI datapaths.

sEMG robotic hand — Built an embedded human-interface project using surface EMG; later described in *Circuit Cellar*.

Pipe-frequency analysis — Applied DOLFINx-based finite-element analysis to steel-pipe frequency studies for well-tool contexts.

Education

PhD, Electrical Engineering

Yale University

2021–present

MEng and BS, Electrical and Computer Engineering

Cornell University,

2013–2018

Additional Background

Research computing: Python, C++, MPI, OpenMP, CUDA, ROCm, PETSc, SLEPc, Frontier, Perlmutter